

PROVENANCE AND EVIDENCE SUPPLEMENT FOR HIGH-WINDING STATIONARY SOLUTIONS AND SINGULAR EXITS IN THE VAN DER POL REACTION-DIFFUSION SYSTEM

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1. PURPOSE AND RELEASE IDENTITY

This supplement records evidence status, source identities, theorem dependencies, and replay information for the companion paper. It is not a second mathematical proof and does not enlarge the theorem. Two immutable public records have different roles in the argument. The publicly archived fixed-system preprint is Zenodo version 1.0.0,

<https://doi.org/10.5281/zenodo.22139860>,

whereas the version-1 application proof dossiers, certificates, and exact frozen long-draft snapshot are fixed together by

[https://github.com/h-lu/rfsn-ii-positive-parameter-pde/
releases/tag/vdp-companion-v1](https://github.com/h-lu/rfsn-ii-positive-parameter-pde/releases/tag/vdp-companion-v1).

The two records are not interchangeable. The Zenodo paper is the stable source for the fixed-Hamiltonian high-winding theorem and its RFSN-II realization. The application release is the immutable evidence baseline for the older long-draft clauses not retained in that focused paper and for the coordinate-level version-1 positive-parameter arguments. The present reader-facing manuscript may be revised without changing that archived baseline. The read-only upstream repository was not modified.

The companion theorem is conditional on Hypothesis H. Its analytic application conclusions are existential on a nonempty compact positive annulus. The proposed numerical Issue #7 box is not used as a theorem box.

2. PUBLIC FIXED-SYSTEM SOURCE AND FROZEN LONG SOURCE

2.1. Public fixed-system preprint. The stable fixed-system source is H. Lu, *High-winding first-hit geometry and singular action finite parts near a reversible Hamiltonian saddle-focus*, Zenodo version 1.0.0, DOI <https://doi.org/10.5281/zenodo.22139860>. Its source revision is

3516a2ee499fc74a46d0ec204736bde8d3782c59,

and the deposited manuscript PDF has SHA-256

569906de86c7d3a716831e036056cfeebb772d5a87fb89c04ee61ee3daf0787d.

Date: September 1, 2026.

That record contains the focused manuscript, its self-contained source, and the M1–M4 evidence used for the RFSN-II realization. In particular, it is the stable source for the fixed-system first-hit theorem, the RFSN-II core homoclinic and three selected channels, the fixed-system return and action modules, and the countable coding consequence; the principal locations are Theorems 2.9 and 6.3 and Corollaries 7.1–7.2. Its public replay summary records **PASS** for the M1–M4 **main-theorem** profile. This is a replay of the stated computer-assisted inputs, not an independent review of the analytic arguments.

The focused paper deliberately treats one fixed Hamiltonian. It does not state the compact-parameter, mixed state/parameter versions used by the application, and it does not contain the optional Jost module described below. Moreover, its concrete pole theorem concerns an isolated boundary sink with indicial roots $\{1, 4, 6\}$; it is not the positive-parameter van der Pol pole theorem.

2.2. Frozen long research version. The auxiliary frozen source is the earlier long research version H. Lu, *First returns, singular exits, and action finite parts near a reversible Hamiltonian saddle-focus*, originally stored in the historical repository **h-lu/reversible-rfsn-ii-waves**. It is not another name for the Zenodo paper. The inspectable copy is the exact snapshot inside the application release. Its upstream commit is

`d54add098545063d5efe8f1d6f062d4cfc116a0d`

and tree

`b9cd34b1bae1c29bdd722d9ed3c33402c4b3ee89.`

The public application release contains an exact **git archive** extraction of the complete frozen **papers/paper-a/manuscript/** and **validation/** directories at

`frozen-imports/rfsn-ii-d54add098545063d5efe8f1d6f062d4cfc116a0d/
source/.`

No file below that directory was edited after extraction.

The principal SHA-256 identities of this long snapshot are:

frozen file	SHA-256
manuscript source	0baf6335aad72d5893479d8876d2613671ecb8ac2ccd73664405dea4381e6a20
manuscript PDF	67888cf8b61b34c923cf55bd69ee41cab69493be6fc275533c8f5a074f1e96c5
replay manifest	15905f5a20b24a0ae0d298d9d14aa940177a006cc2cce3f2a39fd2e1cf4dac9b
environment lock	6240ebdbf0f296738534c07a33aea40202883f6abf37e1ff28e43dad47aa0cba
core homoclinic certificate	ed0f9f58f8ba5f1d5c36dc7c3a72bb725599c4172a3cd610d890b88699fecfbd
algebraic-anchor certificate	60882ee1d3b2b18264b85764288505ae8b47d00bc826a2bddec152898f690fbe
future-target certificate	88fa64035bb4352f5e25aa8d1627b191936264958c125dface59c5a767f6b3ce
pole-gate certificate	7f325a87810f8b0dda2542aed90263b39f9a9f4bd2e8ebb3abcc238d032eb6e2

The machine-readable copy of this table is **RELEASE_MANIFEST.json** beside the frozen snapshot **README**.

The four displayed claim-bearing certificate files are byte-identical to the corresponding M1–M4 evidence in the Zenodo record. The long snapshot

is kept because some analytic clauses used by the companion are absent from, or strictly stronger than, the focused paper; it is not used to replace the stable citation when the focused theorem already supplies the required fixed-system statement.

3. IMPORTED CLAUSES AND EXCLUSIONS

The following clauses of Hypothesis H are supplied directly by the public fixed-system preprint.

- (1) The core system, one selected symmetric homoclinic, transversality modulo flow in the regular zero-energy hypersurface, its true unstable source graph, the certified channel anchors, and the adapted finite central physical event block.
- (2) The fixed-system saddle passage, winding selector, whole-cell convergence, clean first-event decomposition, exact finite-branch action and covariance, and countable coding of the trapped return set.
- (3) The C^3 , third-order-bunched singular algebraic comparison sheet and the validated transverse source-to-sheet intersection. The nonzero source-incidence row in Hypothesis H is the corresponding section-level restatement of this bordered transversality.

Only the following additional clauses are taken from the frozen long version.

- (1) The C^4 , fourth-rate-gap form of the canonical singular algebraic comparison sheet used in the matching proof (Proposition 8.1 of the long source).
- (2) The exact Jost basis and tangent identification, with the endpoint-normalized adjoint row whose directed exchange pairing is $144\sqrt{3}$. This coefficient is not a theorem statement in Zenodo version 1.0.0; it is the explicit algebraic consequence of the frozen Cauchy data and symplectic pairings in `app:jost-source-arm`, as recorded in the application proof archive.
- (3) The compact-parameter architecture for the endpoint and matching equations, mixed passage and selector, mixed two-jet estimates, and finite-atlas covariance. The concrete compact-family hypotheses and the finite marked-atlas descent are checked in the companion paper, with coordinate-level details retained in the application archive; the long version does not by itself verify their van der Pol hypotheses.

Thus Hypothesis H combines a publicly archived fixed-system theorem with a small, explicitly identified set of long-version clauses. Its conditional status is a theorem-scope statement, not a claim that the focused flagship preprint is unarchived or inaccessible.

The following conclusions are not imported from either source:

- either concrete singular end as a positive-parameter van der Pol end;
- a positive-parameter pole or algebraic compactification;
- a positive-parameter $K_2 \rightarrow K_1 \rightarrow$ outer matching theorem;
- a positive-parameter terminal counterterm or finite part;

- a global exact saddle chart or one parameter-global winding alphabet;
- temporal PDE stability, temporal chaos, Turing selection, canard identification, or experiment.

Thus the two positive ends, matching bridge, finite parts, event refinements, and translation to the original PDE are application results, not imported conclusions of the fixed-system theory.

4. CLAIM REGISTER CROSSWALK

The status column below reproduces the authoritative register verbatim. The dependency column records where a proved application statement uses Hypothesis H or an earlier H-dependent statement. Thus *Proved* is a proof status, not a claim that the frozen certificates were independently replayed by this repository.

ID	statement	status	dependency
V1	physical reversible exact Hamiltonian structure and primitive	Derived	direct calculation
V2	central saddle-focus, selected homoclinic, exact passage, and finite events	Proved	public H_{core} ; long/local family passage
V3	genuine positive pole, open basin, labels, and pole action finite part	Proved	V2 pole window
V4	positive outer future-staying hypersurface and algebraic asymptotics	Proved	direct full-PDE construction
V5	$K_2 \rightarrow K_1 \rightarrow$ outer matching and moving-cut covariance	Proved	V2, V4; public incidence and long Jost row
V5A	reference-subtracted algebraic length/action finite parts	Proved	matched V5 arrival
V6	exhaustive physical high-winding return/first-exit relation	Proved	V2–V5A; public fixed-system and long/local family modules
V7	periodic, multipulse, and aperiodic stationary PDE patterns	Proved	V6; public/local coding modules

ID	statement	status	dependency
V8	periodic centered-moment criterion for a positive co-periodic eigenvalue	Proved	direct temporal operator-pencil argument
V8A	frozen A2 true profile satisfies the V8 criterion	Local mathematical PASS; non-claim-bearing	independent replay pending
V9	localized centered-moment criterion for a positive L^2 eigenvalue	Proved	direct temporal operator-pencil argument
V9A	frozen <code>pulse_1</code> true profile satisfies the V9 criterion	Local mathematical PASS; non-claim-bearing	independent replay pending
V10	exact exclusion of classical stationary Turing onset and frozen-box separation	Proved	direct Fourier-symbol calculation
V11	a real positive temporal eigenvalue implies nonlinear orbital instability	Proved	semilinear time-map argument
V11A	frozen A2 co-periodic nonlinear orbital instability	Local mathematical PASS; non-claim-bearing	V8A; replay pending
V11B	frozen <code>pulse_1</code> whole-line nonlinear orbital instability	Local mathematical PASS; non-claim-bearing	V9A; replay pending
T2G	one global exact saddle chart and global winding alphabet	Proposed	not used
S1	temporal stability of a B2 or V7 profile	Deferred	complete spectral exclusion and nonlinear stability argument
E1	calibrated experimental observation	Deferred	separate problem

5. APPLICATION PROOF LOCATIONS

The reader-facing paper compresses the proof into four mathematical sections. The complete coordinate-level arguments remain in the versioned archive:

output	repository proof file
physical model, clocks, Hamiltonian	van-der-pol/HAMILTONIAN_CHECK.md;
frozen core statement	van-der-pol/MODEL_AND_CENTRAL_CHART.md
central continuation	van-der-pol/CENTRAL_CORE_IMPORT.md
positive pole terminal	van-der-pol/CENTRAL_CONTINUATION.md
outer future-staying graph	van-der-pol/POSITIVE_POLE_FINITE_PART.md
relative corner invariant manifold	van-der-pol/OUTER_FUTURE_STAYING.md
central-outer matching	theory/RELATIVE_OVERFLOWING_NHIM.md
representative finite- K_1 /source-incidence check	van-der-pol/CENTRAL_OUTER_MATCHING.md
algebraic finite parts	validation/rigorous/V5_REPRESENTATIVE_BRIDGE_REPORT.md
frozen return/coding statement	van-der-pol/OUTER_ALGEBRAIC_FINITE_PART.md
conditional compact-family transfer	van-der-pol/RETURN_EXIT_CODING_IMPORT.md
physical finite-atlas descent	theory/COMPACT_FAMILY_FIRST_HIT_THEOREM.md
first events, actions, and PDE coding	theory/FINITE_MARKED_ATLAS_DESCENT.md
periodic spectral criterion and A2 validation	van-der-pol/TWO_END_RETURN_EXIT_AND_PDE.md
localized spectral criterion and pulse_1 validation	van-der-pol/A2_PERIODIC_SPECTRAL_INSTABILITY.md; validation/a2_periodic/README.md
classical stationary Turing exclusion	van-der-pol/PULSE_1_SPECTRAL_INSTABILITY.md; validation/pulse_1/README.md
nonlinear orbital-instability bridge	van-der-pol/CLASSICAL_STATIONARY_TURING_EXCLUSION.md
arrow-by-arrow proof audit	van-der-pol/NONLINEAR_ORBITAL_INSTABILITY.md
	proof-audit/VDP_PUBLICATION_PROOF_AUDIT_2026-08-28.md

6. REPRESENTATIVE INTERVAL CHECK OF THE DECISIVE BRIDGE

The analytic proof in the main paper separates two facts: the resolved K_1 construction transports the outer future-staying sheet to a central target graph, and the phase-time incidence proves that a true source orbit meets that graph. Neither fact alone produces the selected algebraic orbit. The application archive now contains a rigorous interval check of the same composition on one explicit parameter cell. This check is supplementary; the existential annulus in the main theorem does not depend on it.

Let

$$\mathcal{C}_* = \left[\frac{96}{6400}, \frac{97}{6400} \right] \times \left[0, \frac{1}{256} \right] \times \left[1, \frac{121}{120} \right].$$

For every $\mu \in \mathcal{C}_*$, the complete-v2 finite- K_1 proof supplies the actual C^3 pullback of the V4 terminal graph on the $U = -4$ section, written

$$n = g_\mu^-(b), \quad |b| \leq \frac{27}{200000}, \quad |g_\mu^-(b)| < \frac{1}{12500}, \quad \text{Lip}(g_\mu^-) \leq \frac{7}{10}.$$

The exact K_1 -to-central map regraphs this curve regularly in a fixed physical patch. Independently, the representative-cell incidence certificate proves, uniformly in μ , that the canonical true source meets every graph satisfying these three bounds exactly once in the specified source-phase component. Instantiating that universal statement with g_μ^- gives

$$\forall \mu \in \mathcal{C}_* \quad \exists! \theta_\mu \quad \text{such that} \quad S_\mu(\theta_\mu) \in \text{graph } g_\mu^-.$$

The intersection is transverse, and its forward orbit crosses the finite K_1 tube to the actual V4 future-staying graph. No parameter regularity of θ_μ is inferred from this interval composition.

Calling S_μ the canonical true source also uses the P2b0 true-graph error certificate, the P2bK positive-Kato source parameterization, and the P2e exact zero-energy source chart. Those imports and their hashes are listed in the composition report; the incidence executable alone proves a kernel for the displayed source-tube contract and does not authenticate the source identity by itself.

All inputs have local mathematical status **PASS** on their stated domains and retain **claim_bearing=false**; the required independent replay has not been completed. The statement is only on $\mathcal{C}_*$. The lower, centre, and upper grouped cells do not interpolate to a cover, and no conclusion is drawn on the untested cells of the frozen v2 box. The 393,216-cell extension is deliberately deferred to [Issue #14](#). It is not a completion condition for this paper.

7. REPLAY INSTRUCTIONS AND EVIDENCE LEVEL

Two replay statements must be kept distinct. The Zenodo archive includes a public replay summary with status **PASS** for the fixed-system paper's M1–M4 **main-theorem** profile. That replay concerns the four computer-assisted RFSN-II inputs only. It neither validates the companion's positive annulus nor constitutes an independent review of the proofs.

In a full Git clone of the application repository checked out at the release tag, change to the frozen **source/** directory and run

```
python3 validation/replay_all.py \
  --profile main-theorem \
  --dry-run \
  --report /tmp/rfsn-main-theorem-dry-run.json
```

checks the manifest and frozen input hashes. Its successful status is

PASS-DRY-RUN-NO-CERTIFICATES-EXECUTED.

By design, this is not a mathematical replay. The frozen script is fail-closed and invokes Git, so this command is not advertised for a GitHub source ZIP or for a standalone copy without **.git**. Its report field **git_commit** identifies the enclosing application checkout; the upstream frozen identity is the **upstream_commit** in **RELEASE_MANIFEST.json**.

A full replay uses the same command without **-dry-run**, supplies the CAPD source and **capd-config** paths, and follows the compiler and FILIB

constraints in the frozen environment lock. The interval computations use outward-rounded CAPD/FILIB enclosures; see the frozen validation README and the software records in the source package. The package is included so that the source claim can be inspected and replayed. This release does not claim that a second machine replay was performed by the application project.

The application repository also contains an unfinished explicit positive-box validation programme under `validation/rigorous/`. Its representative finite- K_1 /incidence composition is described above, but its complete-box aggregate status remains `INCONCLUSIVE`, `claim_bearing=false`. Its development environment lock is distinct from the frozen core environment above and is not evidence for the existential annulus in the paper.

8. REVIEW AND LICENSING STATUS

The companion was subjected to mathematical seam audits, rendered-PDF checks, and a separate AI cold-reading task. These checks are not a substitute for an independent human expert review. A neutral expert packet is included beside the manuscript; no human reading is claimed until a named reader has actually returned a report.

The Zenodo fixed-system deposit contains its own paper and code license files. Those licenses do not retroactively license the distinct application repository or its frozen long snapshot. Neither of the latter contained a license file at the time of the application release; that snapshot therefore preserves inspection and citation access without inventing reuse rights. A future license decision for the application materials belongs to the repository owner and is not a mathematical claim of this supplement.

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